

7 Oct, 2025

Byte Addressable, Word Aligned
Memory

non-conditional $\leftarrow$ Function call
Branch $\rightarrow$ called in ARM

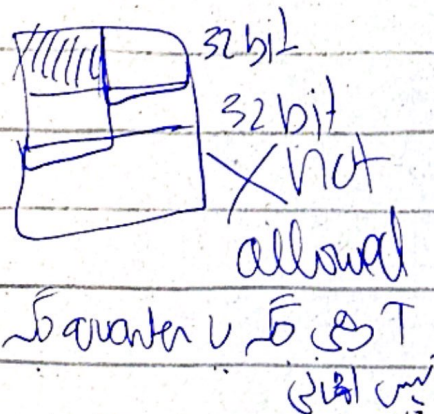
in RISC - V - fixed 20 stores 0
in ARM R0 available.

Stack $\rightarrow$ LIFO

$\hookrightarrow$ temporary memory

Default MSP used $\rightarrow$ for processor

User code/process $\rightarrow$ PSP



bytes
ALIGN $\overline{4}$

ALIGN $\overline{3}$
 $\downarrow$ 3
2
power of 2

DCB $\rightarrow$ 8 bit (char)
DCW $\rightarrow$ 16 bit (short)
DD $\rightarrow$ 32-bit (long)

MY_VAR DCW 0x1234

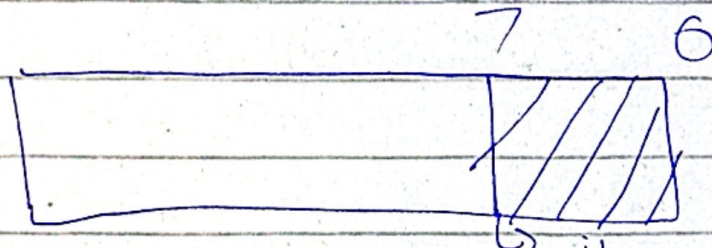
MY_VAR DCW 0x0001, 0x0200

base-address of array

32 bit
MOV $\rightarrow$ lower-bits 16 bit
upper 16 zeros out
MOV T $\rightarrow$ upper-16bits lower 16 zeros

MOV R5 0x00001234
MOVT R5 0xABCD0000

LDR SB



if 0 all 0s filled
LDR SB if 1 all 1s filled

[R0] → treat value in R0
as an address

[R0, #4]

offset
add 4